Priyanshu Srivastava

07352667473 • sri.priyanshu2000@gmail.com • LinkedIn • github.com/priyanshusrivastava2000

SUMMARY

Artificial Intelligence (AI) and Machine Learning (ML) Engineer with 3 years of industry expertise deploying machine learning, deep learning, and reinforcement learning (RL) systems in production, with research spanning NLP, computer vision, and LLM interpretability. Proven background building LLM powered microservices, agentic workflows, and Retrieval Augmented Generation (RAG) pipelines with continuous integration (CI/CD), quantitative benchmarking, and AI observability on cloud native GCP infrastructure — leveraging Python, PyTorch, LangChain, LangGraph, and vector databases, with close collaboration and communication with product and engineering stakeholders.

TECHNICAL SKILLS

Languages: Python • C++ • TypeScript • Java • C#
ML / AI: PyTorch • TensorFlow • scikit-learn • XGBoost • Machine Learning (ML) • Deep Learning (DL) • Neural Networks • Transformers • Reinforcement Learning (RL) • RLHF
LLM Frameworks: LangChain • LangGraph • LlamaIndex • Hugging Face • LLMs • Generative AI • RAG • Agentic AI • Prompt Engineering • Multi Agent Systems
Cloud & Infrastructure: Docker • Kubernetes • Google Cloud Platform (GCP) • FastAPI • Microservices • GPU Training
Data & Search: ChromaDB • Neo4j • Vector Databases • Vector Embeddings • chunking • reranking • hybrid search
Practices: Agile • Distributed Systems • Asynchronous Programming • CI/CD • Model Deployment • Model Monitoring • AI Observability • Benchmarking • pytest • Git

EXPERIENCE

Infosys Limited — **July 2024 – Sept 2025**
Senior AI Engineer — Machine Learning, LLM Agentic Systems and Production RAG Platform — *Bangalore, India*

- Architected and shipped production microservices for model training, inference, and evaluation pipelines with CI/CD automation, applying reinforcement learning (RL), transformer based Hugging Face models, embeddings, and vector databases across 3 distributed Kubernetes GCP environments, reducing manual investigation time by 25 percent daily
- Built and deployed an LLM powered agentic assistant with tool calling, structured outputs, and prompt engineering, integrating a Retrieval Augmented Generation (RAG) backend with chunking, reranking, and hybrid search, enabling autonomous decision making and agent communication for 1000+ customers
- Drove cross functional collaboration with product and engineering stakeholders to design ML driven validation pipelines with automated testing and workflow automation, improving data quality by 15 percent and accelerating ticket processing by 22 percent across 5000+ monthly workflows
- Implemented experiment tracking, model monitoring, AI observability, evaluation harnesses, and quantitative benchmarking across all 6 LLM pipeline components, achieving 40 percent reduction in time to deploy and improving engineering team productivity

Infosys Limited — **July 2023 – July 2024**
Systems Engineer — eBay Platform — *Bangalore, India*

- Managed 200+ production incidents per quarter and deployed automated alerting across 3 engineering teams, reducing incident recurrence by 20 percent and achieving 12 percent faster resolution time

PROJECTS

HealthRAG — Medical Question Answering RAG Agent *Python • LangChain • ChromaDB • Neo4j • FastAPI • Streamlit • Docker • Google Gemini API* — GitHub

- Designed a multi tool LangChain RAG agent with 3+ specialist tools (ChromaDB vector store retrieval, Neo4j graph queries, wait time lookups), demonstrating agentic orchestration and autonomous question answering
- Implemented FastAPI asynchronous backend with Streamlit frontend, Docker containerization, and a pytest suite validating agent tool selection across 5+ entity domains (hospitals, physicians, payers, reviews)

Sparse Autoencoder Probing — LLM Interpretability Research *Python • PyTorch • scikit-learn • XGBoost • Hugging Face • Claude API* — GitHub

- Researched sparse autoencoder (SAE) interpretability for large language models by extracting latent features from Gemma residual stream activations and training cross validated classifiers (logistic regression, XGBoost, MLP) across 160+ binary classification datasets
- Evaluated feature quality under 4+ experimental settings (data scarcity, class imbalance, label noise) with separability and selectivity analysis, extending SAE text latents to ResNet50 CIFAR 100 computer vision features

EDUCATION

University of Southampton — **Sept 2025 – Present**
MSc Computer Science — *Southampton, United Kingdom*

- Researched Reinforcement Learning (policy gradient, reward shaping, alignment), Machine Learning, NLP, neural networks, and Deep Learning; designed training environments and evaluation methodologies across 4+ research projects

Shri Mata Vaishno Devi University — **Aug 2018 – July 2022**
BTech Computer Science and Engineering, Jammu, India — CGPA: 7.93 • Top 15 percent of 120 engineers